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IB Mathematics Application & Interpretation
Higher Level–Prediction Exams
November 2026 - Paper 1

Duration: 2 hrs **(Calculator is Allowed)** **Max. Marks: 110**

Question 1

[Maximum mark: 6]

Amina deposits \$60,000 at a nominal annual rate of 4.80% compounded quarterly. After 4 years the balance is $\$72,617.19$

(a) Verify the balance.

She then invests the balance at 4.20% nominal interest compounded monthly and withdraws \$500 at the end of each month.

(b) Find the balance after 10 years, to the nearest dollar.

(c) State whether the account is exhausted by then.

Answer: Using GDC

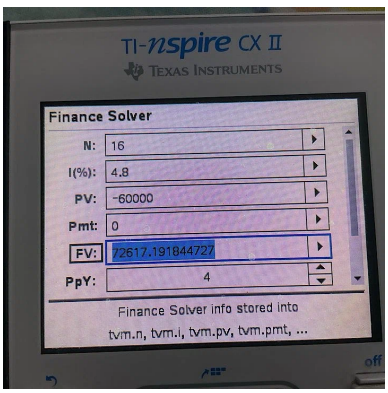
N	K	I	PV	Pmt	FV	Ppy	Cpy
4	4	4.80	−60,000	0	—	4	4

$$FV = PV\left(1+\frac{r}{100k}\right)^{nk}$$

$$= 60000\left(1+\frac{4.80}{400}\right)^{4\times 4}$$

$$FV \approx \$72617.19$$

Balance is verified



(b)

N	I	PV	Pmt	FV	P_{py}	C_{py}
12x10	4.20	−72,617.19	500	—	12	12

Using GDC

FV = $ 36,033.00

(C) As the amount after 10 years is still positive, so account is not exhausted.

Question 2

[Maximum mark: 6]

A biologist is studying the changing size of a bacteria culture in a laboratory experiment.

The number of bacteria, N, can be modelled by the function $N = 800 + 200(1.15)^{-t}, \quad t \geq 0$ where t is the number of hours since the start of the experiment.

(a) State the number of bacteria at the start of the experiment. [1]

(b) Determine the number of whole hours it takes for the bacteria population to decrease by 15%. [3]

(c) Determine the smallest possible number of bacteria in the culture, according to this model. Explain what this value represents in the context of the experiment. [2]

Answer:

(a) $N(0) = 800 + 200(1.15)^{0} = \boxed{1000}$

(b) 15% of 1000 = 150 decreased

$$\therefore N = 850$$

$N = 800 + 200(1.15)^{-t}$

$$\therefore 850 = 800 + 200\,(1.15)^{-t}$$

using GDC

$$t = 9.9189\ldots \approx \boxed{10} \text{ Hours}$$

(c) As $t \to \infty, (1.15)^{-t} \to 0$

$$\therefore N \to \boxed{800}$$

Question 3

[Maximum mark: 6]

40 students were asked to volunteer picking up rubbish at a local beach. The amount of time each student spent picking up rubbish was recorded and counted towards their school service record. One student did not arrive and was recorded as 0 minutes.

One student spent 60 minutes collecting rubbish, more time than any other student. The second most time spent by any student was 50 minutes.

The median time for the 40 students was 30 minutes.

25% of students spent 20 minutes or less picking up rubbish.

(a) Given that the inter-quartile range was 14 minutes, determine the upper quartile. [1]

Answer:

(a) $IQR = Q_3 - Q_1$, $Q_2 = 30$, $Q_1 = 20$

$$Q_3 - Q_1 = 14$$

$$\therefore Q_3 - 20 = 14 \Rightarrow Q_3 = \boxed{34}$$

(b) Hence, determine whether the student who spent 60 minutes picking up rubbish was an outlier. [2]

Upper boundary for outlier is,

$$34 + 1.5\,(IQR) = 34 + 1.5 \times 14 = 55$$

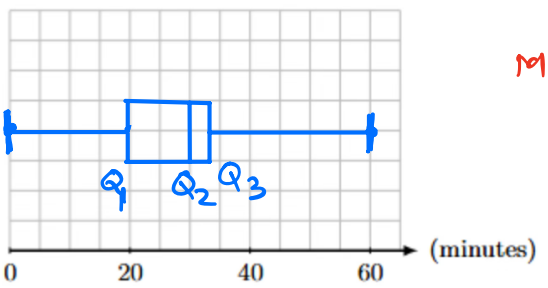
Since $60 > 55$

The student was an outlier



(c) Construct a box and whisker diagram for the times of the 40 students on the following axes. [3]

5-point summary
Min, Q_1, Q_2, Q_3, Max.



Question 4

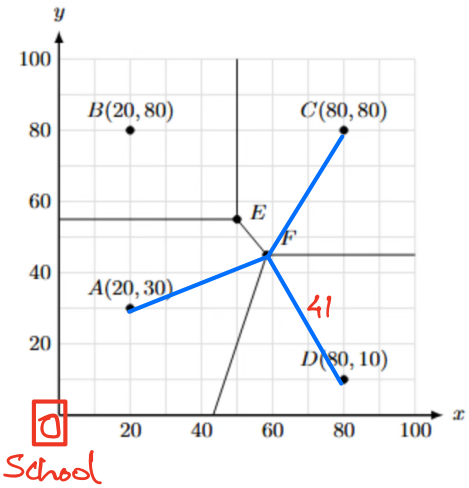
[Maximum mark: 6]

A school fair has been set up on a field that is 100 metres long and 100 metres wide. There are to be five stalls at the fair. When Elias arrives to set up his stall, the other four stalls are already set up.

Ali has set up his stall at point A, 20 metres east and 30 metres north of a school building at point $(0, 0)$. Brenda has set her stall up at point $B(20, 80)$, Clem has set his stall up at point $C(80, 80)$ and Dana has set her stall up at point $D(80, 10)$, all relative to the school building at the origin.

The Voronoi diagram below shows the stall locations. Points E and F are at the two corners of the Voronoi diagram inside the school field.

east → x
North → y



(a) Find the coordinates of point E. [2]

Answer: Point E is horizontally in the middle of point A and B, and vertically in the middle of points B and C.

Therefore, x coordinate of E is $x = \frac{20 + 80}{2} = 50$

And y coordinate of E is $y = \frac{30 + 80}{2} = 55$

$\therefore E\,(50, 55)$

It is given that point F is 41 metres away from Dana's stall, to the nearest metre.

Elias wants to set up his stall so that it is as far away from the other four stalls as possible.

(b) Verify that Elias should set up his stall at point F. [4]

Answer: Since F is the vertex of Voronoi diagram,

$\therefore FA = FC = FD = 41$

A (20, 30) ✓
E (50, 55) ✓

Now distance EA is

$$EA = \sqrt{(50 - 20)^2 + (55 - 30)^2} \approx 39.1m$$

Since 41 > 39.1, F is further from its closest stalls , A, C and D than E is from its closest stalls A, B and C.

Elias should set up his stall at the point F.

Question 5

[Maximum mark: 6]

A researcher at a school wants to investigate whether there is an association between a student's preferred method of revision and their year group. A random sample of 250 students was surveyed, and the results are shown in the following table.

	Year 12	Year 13	Total
Flashcards	40	35	75
Practice questions	55	60	115
Video tutorials	25	35	60
Total	120	130	250

A χ^2 test for independence is to be carried out at the 5% significance level.

(a) (i) State the null and alternative hypotheses for this test.

(ii) State the number of degrees of freedom for the test.

(iii) Find the χ^2 statistic for the test. [4]

Answer:

(a) (i) H_0 : Student's preferred revision method is independent of age group.

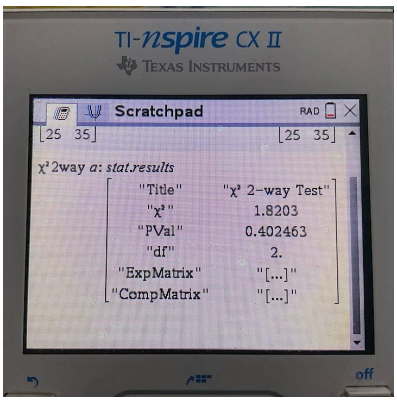
H_1 : Student's preferred revision method is not independent of age group.

(ii) Degree of freedom: $(3-1)(2-1) = 2$

(iii) Using GDC, χ^2 statistics is

$$\chi^2 = 1.82$$

Matrix → menu → Matrix → ...



The χ^2 critical value for this test is 5.991.

(b) State the conclusion to the test. Justify your answer. [2]

Since $\chi^2_{cal} < \chi^2_{crit}$

i.e. $1.82 < 5.991$

( )!

Ctrl+Var → a

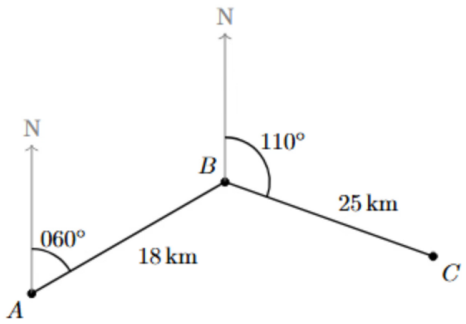
Therefore we do not reject the null hypothesis.

Hence there is insufficient evidence to suggest an association between revision method and year group.

Question 6

[Maximum mark: 7]

A ship sails from port A for 18 km on a bearing of 060° to reach point B. It then changes course and sails 25 km on a bearing of 110° to reach point C. This is shown in the following diagram.



(a) Find the distance AC. [3]

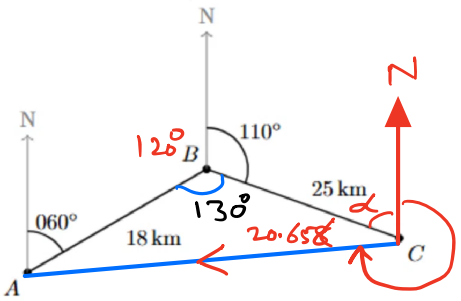
The ship will sail in a direct line from point C back to the port at point A.

(b) Calculate the bearing the ship will sail on from C to A. [4]

Answer:

$(a)\ A\hat{B}N = 180 - 60 = 120°$

$\therefore A\hat{B}C = 360 - 120 - 110 = \boxed{130°}$



Using cosine rule,

$$AC^2 = 18^2 + 25^2 - 2(18)(25)\cos 130°$$
$$= 1527.5088\ldots$$

$$\therefore AC = 39.1km$$

$a^2 = b^2 + c^2 - 2bc\cos A$

(b) $\alpha = 180 - 110 = 070°$

Using sine rule,

$$\frac{\sin B\hat{C}A}{18} = \frac{\sin 130°}{39.0833577}$$

$$\therefore Sin B\hat{C}A = 0.3528\ldots$$

$$\therefore B\hat{C}A = 20.658$$

Therefore bearing of A from C is

$$\text{bearing} = 360 - 70 - 20.658 = 269.3\ldots = 269°$$

$\therefore$ Bearing of A from C is 269°

Question 7

[Maximum mark: 7]

A pharmacy sells shampoo by the bottle. The volumes of the shampoo bottles are known to be normally distributed with a mean of a millilitres and a variance of 16 millilitres. They also sell bottles of conditioner, the volumes of which are normally distributed with a mean volume of b millilitres and a variance of 9 millilitres.

The pharmacy sells the shampoo and conditioner in a "value pack" containing one bottle of shampoo and one bottle of conditioner.

(a) State the distribution of the "value pack", and, in terms of a and b, the mean and variance. [3]

Answer: Let V represents value pack, S represents Shampoo and C represents Conditioner.

$$\therefore V = S + C$$

Since S and C both follows normal Distribution, V also follows a normal Distribution.

Using linear combination of independent random variables (4.14)

$$E(V) = E(S + C)$$
$$= E(S) + E(C)$$
$$\therefore E(V) = a + b$$

And since S and C are independent

$$Var(V) = Var(S + C)$$
$$= Var(S) + Var(C)$$
$$\therefore Var(V) = 16 + 9 = 25 = 5^2$$

$$\therefore V \sim N(a + b, 5^2)$$

$\sigma^2 = 5^2$

$\sigma = 5$

A sample of 10 value packs is found to have a mean volume of 997 millilitres.

(b) Determine a 95% confidence interval for the mean volume of the value pack. Give your answer to 5 significant figures. [2]

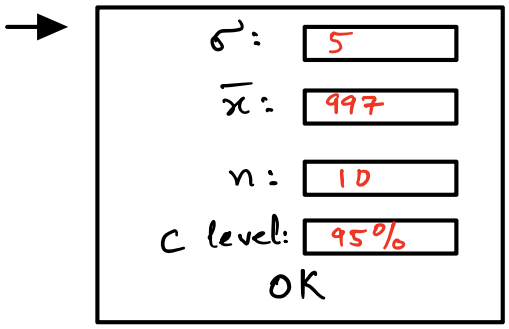
(c) State, with a reason, whether the pharmacy is justified in advertising the value pack with a total volume of 1 litre. [2]

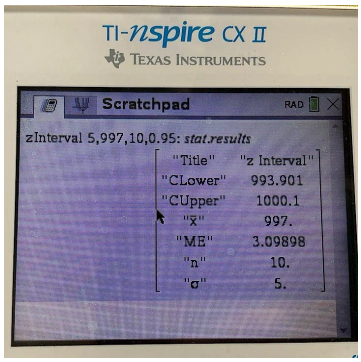
Answer:

(b) since the population variance is known, using GDC to construct 95% z–interval

$(993.90, 1000.1)$ (5 s. f.)

Menu → $stat$ → 6 Confidence interval → Z– interval → Data: stats





(c) Since the total volume of 1 litre (1000 ml) lies within the confidence interval (993.90, 1000.1) found in part (b)

Yes the claim is justified

Question 8

[Maximum mark: 7]

The height of a mountain bike trail can be modelled by the function

$$h(x) = \frac{\ln(x+1)}{(x+1)^2}, \quad 0 \le x \le 4,$$

where x is the horizontal distance in kilometres from the start of the trail and h is the vertical height of the trail in kilometres at distance x, relative to the start of the trail at $(0, 0)$.

(a) Show that

$$h''(x) = \frac{6\ln(x+1) - 5}{(x+1)^4}.$$

[4]

(b) Hence determine the x-coordinate of the point at which the trail is descending most rapidly. Give your answer in the form $e^a - b$, where $a, b \in \mathbb{Q}^+$. [3]

Answer: (a)

$$h(x) = \frac{\ln(x+1)}{(x+1)^2}$$

$$h'(x) = \frac{\frac{1}{(x+1)} \cdot (x+1)^2 - \ln(x+1) \cdot 2(x+1)}{(x+1)^4}$$

$$= \frac{(x+1)(1 - 2\ln(x+1))}{(x+1)^{4\,3}} = \frac{(1 - 2\ln(x+1))}{(x+1)^3}$$

$$h''(x) = \frac{-\frac{2}{x+1}(x+1)^{3} - (1 - 2\ln(x+1))\,3(x+1)^2}{(x+1)^6}$$

$$= \frac{(x+1)^2(-2 - 3 + 6\ln(x+1))}{(x+1)^{6\,4}}$$

$$\therefore h''(x) = \frac{6\ln(x+1) - 5}{(x+1)^4}$$

(b) The trail is descending most rapidly at the minimum of h′(x), which occurs when h′′(x) = 0

Using result from part (a),

$$\frac{6\ln(x+1) - 5}{(x+1)^4} = 0$$

$$6\ln(x+1) - 5 = 0$$

$$\ln(x+1) = \frac{5}{6} \Rightarrow x + 1 = e^{5/6} \Rightarrow x = e^{5/6} - 1$$